Diffusion Processes in Brief

SDE, transition densities, Kolmogorov equations, reversibility, time reversal

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The stochastic differential equation

$$\boxed{\mathrm{d}X_t = b(X_t, t)\,\mathrm{d}t + \sigma(t)\,\mathrm{d}W_t} \qquad X_0 \sim p_0, \qquad a(t) = \sigma(t)\sigma(t)^\top \succeq 0.$$

- b: **drift** (mean velocity); σ: noise matrix; a: **diffusion matrix** (covariance per unit time).
- Meaning: $X_t = X_0 + \int_0^t b(X_r, r)\,\mathrm{d}r + \int_0^t \sigma(r)\,\mathrm{d}W_r$.
- Over a short time h (Euler–Maruyama):

$$X_{t+h} - X_t \approx b\,h + \sigma\,(W_{t+h} - W_t): \qquad \mathbb{E}[\cdot \mid X_t] = b\,h, \quad \mathrm{Cov}[\cdot \mid X_t] = a\,h.$$

  Noise is of size $\sqrt{h} \gg h$, so it acts at *second* order.

process	b	σ
Wiener	0	I
Langevin for π	$\nabla \log \pi$	$\sqrt{2}\,I$
OU / DDPM forward	$-\frac{1}{2}\beta(t)x$	$\sqrt{\beta(t)}\,I$

Transition densities and the generator

Transition density $p(y, t \mid x, s)$: density of X_t at y given $X_s = x$. **Chapman–Kolmogorov**:

$$p(y, t \mid x, s) = \int p(z, r \mid x, s)\, p(y, t \mid z, r)\, \mathrm{d}z, \qquad s < r < t.$$

Definition (Kolmogorov): diffusion process

A Markov process with continuous paths such that, as $h \downarrow 0$,

$$\tfrac{1}{h} \int_{|y-x|>\varepsilon} p\, \mathrm{d}y \to 0, \qquad \tfrac{1}{h} \int_{|y-x|\le\varepsilon} (y-x)\, p\, \mathrm{d}y \to b(x,t), \qquad \tfrac{1}{h} \int_{|y-x|\le\varepsilon} (y-x)(y-x)^\top p\, \mathrm{d}y \to a(t),$$

with $p = p(y, t+h \mid x, t)$: no jumps, local drift b, local covariance rate a.

Generator (second-order, elliptic if $a \succ 0$):

$$\mathcal{L}_t f(x) = \lim_{h \downarrow 0} \frac{\mathbb{E}[f(X_{t+h}) \mid X_t = x] - f(x)}{h} = b \cdot \nabla f + \tfrac{1}{2}\, a : \nabla^2 f.$$

For $\sigma = \sqrt{2} I$: $\mathcal{L} = b \cdot \nabla + \Delta$.

The two Kolmogorov equations

Split off the **first** short step of Chapman–Kolmogorov $\Rightarrow$ equation in the *start* (x, s); split off the **last** step and integrate by parts $\Rightarrow$ equation in the *end* (y, t).

$$\textbf{backward:} \quad -\partial_s\, p(y,t \mid x,s) = \mathcal{L}_s\, p \qquad \text{(acting on } x\text{)}, \quad p \to \delta(y-x) \text{ as } s \uparrow t,$$

$$\textbf{forward:} \quad \partial_t\, p(y,t \mid x,s) = \mathcal{L}_t^*\, p \qquad \text{(acting on } y\text{)}, \quad p \to \delta(y-x) \text{ as } t \downarrow s,$$

with the adjoint in divergence form

$$\mathcal{L}_t^*\rho = -\nabla\cdot\big(b\rho\big) + \tfrac{1}{2}a : \nabla^2\rho = -\nabla\cdot J, \qquad J = \underbrace{b\rho}_{\text{transport}} - \underbrace{\tfrac{1}{2}a\nabla\rho}_{\text{spreading}}.$$

Example (Wiener): $p(y,t \mid x,s) = \mathcal{N}(y; x, (t-s)I)$ solves $\partial_t p = \frac{1}{2}\Delta_y p$ and $-\partial_s p = \frac{1}{2}\Delta_x p$.

Marginals and expectations: the two averages

Average over the start

$p_t(y) = \int p_0(x)\, p(y, t \mid x, 0)\, \mathrm{d}x$

$$\partial_t p_t = \mathcal{L}_t^* p_t = -\nabla \cdot \left(b p_t - \tfrac{1}{2} a \nabla p_t\right)$$

Fokker–Planck (forward). Point start $p_0 = \delta_{x_0}$: p_t = transition density.

Average over the end

$u(x, s) = \mathbb{E}[f(X_t) \mid X_s = x] = \int f(y)\, p(y, t \mid x, s)\, \mathrm{d}y$

$$-\partial_s u = \mathcal{L}_s u, \qquad u(\cdot, t) = f$$

backward equation.

Duality (integrate by parts):

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E} f(X_t) = \int f\, \mathcal{L}_t^* p_t = \int (\mathcal{L}_t f)\, p_t = \mathbb{E}\, \mathcal{L}_t f(X_t).$$

- $f = x$: $\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E} X_t = \mathbb{E}\, b(X_t, t)$ (noise does not move the mean);
- $f = xx^\top$: $\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E} X_t X_t^\top = \mathbb{E}[X_t b^\top + b X_t^\top] + a(t)$ (noise feeds the covariance).

Reversible diffusions

Time-homogeneous ($b(x)$, a constant). Stationary π: $\mathcal{L}^*\pi = 0$, i.e. $\nabla \cdot J = 0$.

Definition

The diffusion is **reversible** w.r.t. π if $\mathcal{L}$ is symmetric in $L^2(\pi)$: $\int g\,(\mathcal{L}f)\,\pi = \int f\,(\mathcal{L}g)\,\pi$ for all f, g.

Equivalent:

1. zero flux *pointwise*: $J = b\pi - \frac{1}{2}a\nabla\pi = 0$, i.e. $b = \frac{1}{2}\,a\nabla \log \pi$ (a gradient drift);
2. $\mathcal{L}f = \frac{1}{2\pi}\nabla \cdot (\pi\, a\nabla f)$, $\int g\,\mathcal{L}f\,\pi = -\frac{1}{2}\int \nabla g^\top a \nabla f\,\pi$;
3. detailed balance: $\pi(x)\,p(y, t \mid x, 0) = \pi(y)\,p(x, t \mid y, 0)$;
4. started from π, the paths forward and backward in time have the same law.

Wiener: $\mathcal{L} = \frac{1}{2}\Delta$ is symmetric w.r.t. Lebesgue ($\pi \equiv 1$); heat kernel symmetric in x, y. Reversible, but no stationary *probability*: not ergodic.

Langevin: $a = 2I$, $b = \nabla \log \pi$: reversible. Adding $S\nabla \log \pi$ ($S^\top = -S$) keeps π, breaks symmetry: non-reversible.

Time reversal: the density equation

$Y_s = X_{T-s}$ is again Markov (past and future independent given the present), with $\hat{p}(x, s \mid y, t) = p(y, t \mid x, s)\, p_s(x)/p_t(y)$: **not** time-homogeneous.

Step 1. $\frac{1}{2}a : \nabla^2 p = \nabla \cdot \left(\frac{1}{2}\, p\, a \nabla \log p\right)$ turns Fokker–Planck into a transport equation:

$$\partial_t p_t = -\nabla \cdot \Big(\big(b - \tfrac{1}{2} a \nabla \log p_t\big) p_t \Big).$$

Step 2. For $q_s = p_{T-s}$: $\partial_s q_s = -\partial_t p_t$. Add and subtract $\frac{1}{2}a : \nabla^2 q_s$:

$$\boxed{\partial_s q_s = -\nabla \cdot \left(\hat{b}\, q_s\right) + \tfrac{1}{2}\, a(T-s) : \nabla^2 q_s, \qquad \hat{b} = -b(y, T-s) + a(T-s)\, \nabla \log p_{T-s}(y)}$$

A Fokker–Planck equation with a new drift and the **same** diffusion matrix.

Time reversal: the SDE

Theorem (Anderson 1982; Haussmann & Pardoux 1986)

$$\mathrm{d}Y_s = \Big[-b(Y_s, T-s) + a(T-s)\,\nabla \log p_{T-s}(Y_s) \Big]\,\mathrm{d}s + \sigma(T-s)\,\mathrm{d}\bar{W}_s, \qquad Y_0 \sim p_T.$$

- **Noise unchanged, drift corrected** by the score $a\nabla \log p_t$ (no normalizing constant needed).
- **Reversible and stationary:** $p_t \equiv \pi$, $b = \frac{1}{2} a \nabla \log \pi \Rightarrow \hat{b} = -b + 2b = b$: the same process.
- **Wiener from x_0:** $\nabla \log p_t = -(y - x_0)/t \Rightarrow \hat{b} = -\dfrac{y - x_0}{T - s}$: the Brownian bridge back to x_0.
- **DDPM:** $b = -\frac{1}{2}\beta x$, $a = \beta I \Rightarrow \hat{b} = \frac{1}{2}\beta y + \beta \nabla \log p_{T-s}(y)$; start from $\mathcal{N}(0, I) \approx p_T$, use a learned score.
- **Probability-flow ODE** $\dot{x} = b - \frac{1}{2} a \nabla \log p_t$ (the transport velocity of Step 1): same marginals, deterministic, can be run in either direction.